ANGALAPARAMESHWARI A

EMBEDDED SOFTWARE DEVELOPER

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Professional Summary

Embedded Software Developer with 2 years of hands-on experience in microcontroller-based embedded systems. Strong expertise in low-level programming, device driver development, communication protocol implementation, and system debugging. Experienced in developing industrial wireless products using RF, IR, and Bluetooth technologies. Passionate about building reliable, scalable, and energy-efficient embedded solutions.

Education

Bachelor of Engineering – Electronics and Communication Engineering 2019 – 2023

M.A.M College of Engineering and Technology

Percentage: 84%

HSC 2019

Government Girls Higher Secondary School

Percentage: 78%

SSLC 2017

Government High School

Percentage: 96.6%

Technical Skills

- **Programming Languages:** C, Embedded C, Python
- **Microcontrollers:** 8051 (AT89C51), PIC24FJ64GP205, STM32F446RE, ESP32
- **Communication Protocols:** UART, SPI, I2C, CAN
- **IDEs:** MPLAB X IDE, Keil uVision5, STM32Cube IDE, Espressif-IDE
- **Tools:** Enterprise Architecture (VR-5.0).

Experience

Embedded Software Developer

Innospace Automation Services Pvt. Ltd., Chennai Aug 2024 – Dec 2025

- Developed embedded firmware for industrial remote-control systems using RF, IR, and Bluetooth communication protocols.
- Designed and developed device drivers and modular software components; performed unit-level testing and validation.
- Developed Windows-based Python desktop applications using PyQt and Tkinter for device configuration via serial communication.
- Implemented encrypted import and export of configuration files in the desktop application.
- Performed initial board bring-up, power-on testing, and complete hardware validation.
- Built Bluetooth-based wireless configuration systems enabling real-time parameter tuning.

- Debugged firmware and hardware issues using logic analyzers and oscilloscopes.
- Prepared technical documentation for firmware design, configuration workflow, and production support.

Software Developer Trainee

Test Base Solutions Engineering Pvt. Ltd., Bangalore Jul 2023 – Apr 2024

- Performed detailed embedded software design using Enterprise Architect in compliance with ASPICE (SWE.3).
- Worked extensively with UART, SPI, and I2C communication protocols.
- Interfaced and tested peripherals including ADC, DAC, and PWM modules.
- Developed embedded applications using 8051 and PIC microcontroller platforms.
- Implemented UDS diagnostic frames and transmitted diagnostic data using GUI-based tools.

Projects

Wireless IO System

- Designed and developed an industrial wireless IO system to monitor and control digital and analog inputs and outputs.
- Implemented encrypted wireless data transmission to ensure secure and reliable communication.
- Implemented one-way and two-way wireless data transfer and configured the product via serial and Bluetooth interfaces.
- Designed the system to operate reliably in noisy and harsh industrial environments.
- Improved system robustness through error handling and communication retries.

Dual-Mode Wireless IR & RF Remote Control System

- Designed and developed a wireless remote-control system for industrial crane operations using IR and RF protocols.
- Enabled mode switching between IR (short-range, line-of-sight) and RF (long-range, non-line-of-sight) through a display interface.
- Integrated a menu-driven UI for communication selection and sleep configuration.
- Implemented power optimization techniques to improve battery life.
- Enhanced system reliability and operational flexibility in industrial environments.

Temperature Sensor Monitoring and Simulation System with Data Logging

- Designed and developed a temperature monitoring system using the PIC microcontroller and LM35 sensor.
- Implemented real-time temperature acquisition and display on an LCD.
- Added a simulation mode to generate test temperature data for validation.
- Integrated EEPROM-based data logging for long-term storage and analysis.